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Modeling of 3D geometry uncertainty in Scan-to-BIM automatic indoor reconstruction

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ABSTRACT

The rapidly expanding field of Scan-to-BIM applications highlights the importance of model uncertainty assessment in describing the quality of modeling results. Although there have been recent research advancements in point cloud-based building modeling, there has been limited investigation into accurately analyzing error propagation. This paper estimates the geometry uncertainty in 3D modeling based on a strict application of geodetic stochastic modeling. Statistical uncertainty is incorporated into the building reconstruction process and procedures that enable self-verification within this process are developed. The method can be successfully used to evaluate the dimensional uncertainty of generated BIMs, which is especially important in the field of civil engineering with high accuracy requirements concerning metric quality control. Follow-up research will also consider systematic errors and apply the methods to other 3D point cloud acquisition techniques.

1. Introduction

Building Information Models (BIMs) are digital representations of urban facilities that record all data relevant to a building lifecycle, supporting information creation and exchange [1]. They are widely used in many applications, such as building maintenance and inspection, preservation, or emergency response. Accurate digital models of as-built building interiors provide key information for the maintenance and adaptation of existing facilities. Effective decision support is feasible only when accurate, up-to-date, and complete as-built information about existing buildings is available. At present, however, BIMs often do not exist for many buildings, or they represent the 'as-designed' condition of the object, which may not accurately reflect the current facility state because of differences in the construction and periodical renovations.

Traditional ways of collecting as-built information, such as range finders or total stations, are tedious and time-consuming tasks, especially in complex facilities. In the last decade, increasing relevance was gained by Scan-to-BIM approaches, that provide up-to-date 3D point clouds which serve as input data for the generation of 3D indoor models [2]. Thanks to advances in technology, Scan-to-BIM has become an attractive alternative to the traditional techniques of data acquisition, capable of significantly reducing project costs [3]. A growing number of construction practitioners start to use BIMs, noticing their potential as parametric representations and repositories of accurate as-built information. Facilitated data collection triggered the need for 3D reconstruction methods, with the possibly highest degree of automation. So far, 3D models are often generated for the purpose of architectural and visual representation, which does not fulfill many requirements of more advanced applications [4]. In the field of civil engineering, with high accuracy requirements concerning metric quality control in the range of millimeters, such models are often insufficient. Especially, modeling of building interiors still remains an open challenge, due to environmental constraints causing weak scanning geometries, high amount of clutter, and occlusion [5]. At the same time, there is a great need for a thorough evaluation of indoor as-built 3D models, which impose high geometric resolution and reliably confirmed accuracy.

The digital representation of the real world in scan-to-BIM applications is inherently uncertain. During the modeling phase, unavoidable uncertainties of the measured input data propagate into the accuracy of the generated 3D model. Therefore, the compliant building models cannot be considered error-free. It is also important to recognize, that the accuracy level of measurement results may significantly differ from the obtained accuracy level of the modeling product. Processing the data into a model may lead to the cumulation of errors, but on the other hand, redundancy in the measurements will often improve the final quality of

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Table 1
Level of Accuracy structure definition ([48], specified at 95% confidence level).

Level	Upper Range	Lower Range
LOA10	15 cm	5 cm
LOA20	5 cm	15 mm
LOA30	15 mm	5 mm
LOA40	5 mm	1 mm
LOA50	1 mm	0

the output. The intensively growing scope of 3D model applications, far beyond visualization purposes, leads to the increasing relevance of uncertainty assessment, which indicates the quality of the reconstruction results. Inaccurate 3D models can negatively affect the application to which they are applied, like the reliability of risk monitoring or precision of the reference model. Procedures to estimate the uncertainties associated with a measurement system and modeling method are especially needed in the design phase of the Scan-to-BIM process. Such a priori investigation allows for the prediction of the achievable accuracy of the data processing pipeline which is in development [6]. Furthermore, the estimation procedures may provide knowledge about causes of errors and modeling sub-processes that are sensitive to uncertainties.

Each step of the comprehensive Scan-to-BIM process is associated with inaccuracies that affect the final result. Error propagation starts already from the positioning of the data collection device. Also the choice of the ranging technique, calibration, and environmental conditions during the acquisition process are the influential factors. Moreover, multi-scan registration error effects propagate into the accuracy of the acquired 3D points. So far, most research has been focused on the impact of sensor calibration and point cloud registration [7]. Therefore we focus on the final step of the error propagation chain, which leads from the 3D input data to the generated as-built model. Although each aforementioned factor contributes to the accuracy of the final 3D model, most scientific investigation dedicated to the impact on model accuracy and precision of the modeling phase is barely covered. While there are numerous reconstruction methods proposed in the literature, they still lack a thorough investigation of metric tolerances in engineering practice. Estimation of the reliability level of as-built conditions is especially important for analyzable building models, which has to ensure that planned neighboring elements will fit together after installation on site. Available methods for the quality assessment of the reconstructed 3D models are mostly limited to the comparison of the final product against input data or reference models, e.g. [8–9], without addressing the impact of the modeling process itself. Considering that the majority of efforts in the field of Scan-to-BIM applications are associated with the reconstruction part [10–12], explicit analysis of error propagation for this problem is clearly underrepresented.

The aim of this research is to create a framework for establishing a relationship between the quality of collected 3D point cloud data and the quality of the generated model. In the contribution, we propose an analytical approach for the modeling of 3D geometry uncertainty, based

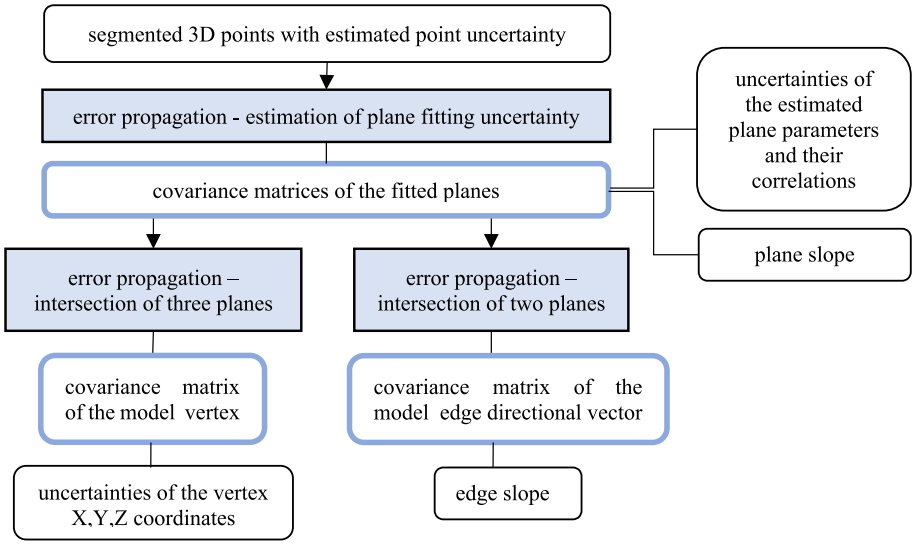

Fig. 1. Overall flowchart of the presented uncertainty estimation methodology.

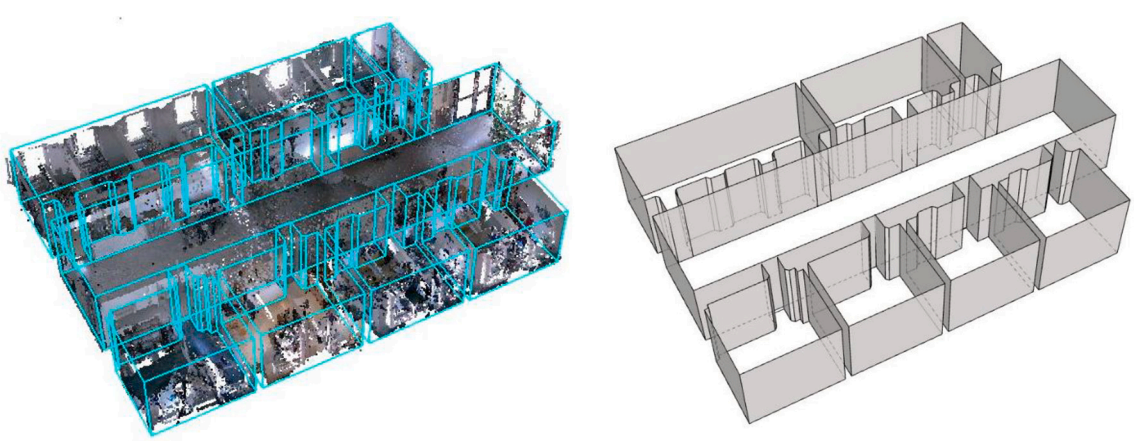

Fig. 2. Input 3D point cloud (left) and the reconstructed 3D indoor model (right).

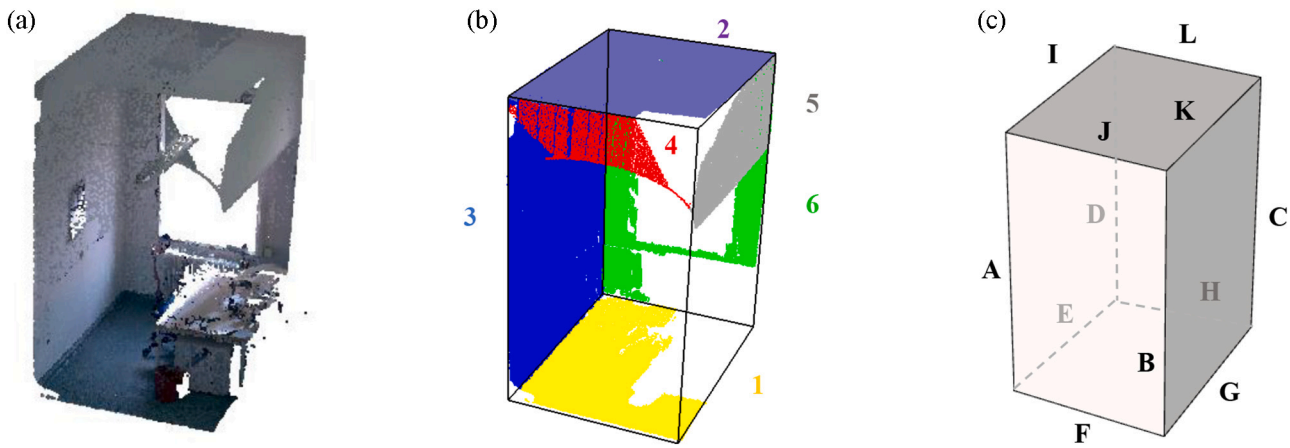

Fig. 3. Exemplary data part (a), direct input to the error propagation analysis – data segmented into planar patches, with plane IDs (b), corresponding vectorised indoor model, with edge IDs (c).

on the first-order Taylor-series expansion. The study focuses on the propagation of uncertainties in fitting objects to 3D points on the positional accuracy of the reconstructed building elements. The new insight of the research comes from considering statistical uncertainties in the building reconstruction process and developing a procedure that allows this process to verify itself. We establish a general model for 3D data fitting, followed by the derivation of the symbolical uncertainty expressions of the reconstructed building elements. All investigations consider the correlations between parameters. Furthermore, we compute confidence intervals to quantify the probability of the reconstructed building elements to belong to certain tolerance thresholds. The implemented procedure enables to integrate the results obtained in terms of error propagation theory with the uncertainty information according to the standardized LOA specification for BIM. The developed theoretical formulae are numerically tested on a 3D point cloud presenting an indoor scene in a university building. In the presented work we are focusing on terrestrial laser scanning (TLS) as data collection technique. Nevertheless, the developed solutions are also relevant for 3D point cloud data acquired in an alternative way, for instance using multi-image-based photogrammetric techniques or through 3D measurement devices in smartphones. Finally, the presented results of the numerical experiment are thoroughly analyzed, quantified, and illustrated.

The structure of this paper is as follows: state-of-the-art developments in the area of scan-to-BIM are described in Section 2. Section 3 presents the methodology of 3D geometry uncertainty modeling. The impact of the reconstruction process on the geometric accuracy of generated building models is presented and evaluated in Section 4. Conclusions are summarized in Section 5.

2. Related work

The use of scan-to-BIM techniques to support building design and construction practices becomes a global standard [13]. The process can be divided into three main steps: (i) data collection and registration, (ii) 3D modeling, and (iii) performance evaluation. Such a frameworks has been adopted as a standard procedure for the generation as-built-BIMs for various purposes, like e.g. indoor navigation [14], emergency response [15], change detection [16], or quality control [17]. Depending on the target application, the requirements for the generated model vary with respect to the geometric accuracy, level of detail, and efficiency of the whole process.

2.1. Data collection and registration

Terrestrial laser scanning techniques are available to quickly collect up-to-date 3D point clouds that can serve as a representation of building interiors. Although constraint indoor conditions affect the quality of TLS data, their degrading impact is smaller as compared to most image-based techniques. Densely collected 3D point clouds enable to present complex geometry, required in construction and civil engineering projects [18]. As with each measurement, TLS data acquisition is characterized by a certain level of accuracy. Errors in data collection propagate through the data processing chain and lead to statistical uncertainty in the final results. A priori knowledge of the stochastic properties of the measurements is therefore important to compute statistically significant geometric attributes of the reconstructed models. Typical errors are connected to the scanning device and the quality of point cloud registration. Complex topics related to errors in TLS have been comprehensively investigated in numerous publications. Error sources are explored together with related influencing parameters and their effects on the geometric quality of collected 3D point clouds in [19,20]. Stochastic models based on intensity information are presented by [21,22]. [23] investigates the magnitude of systematic deviations and estimates measurement uncertainty for the purpose of geodetic applications and deformation analyses. [24,25] model systematic errors and propose calibration strategies to reduce their impact.

2.2. 3D modeling

Scan-to-BIM modeling practice is dominated by manual and semi-manual 3D reconstruction techniques, conducted in BIM authoring software. Automation of 3D reconstruction of as-built models of indoor environments has been addressed in recent years by numerous researchers. Nevertheless, fully automated modeling of arbitrary complex building interiors based on 3D point clouds collected by laser scanners still remains an open challenge [12]. A typical difficulty for indoor scenes is a high amount of clutter and occlusion. Indoor spaces are usually filled with various furniture and other objects, which are difficult to discriminate from building surfaces [26]. Finally, a core reason that makes the process of indoor modeling challenging, is the variety and complexity of the structure of building interiors, which requires generic and flexible reconstruction algorithms. Due to the aforementioned challenges, many approaches were limited to single-room modeling or clutter-free environment, e.g. [27,28]. With the rapid advancement in 3D point cloud acquisition techniques, the need for as-built modeling of a large and complex environment with multiple rooms has come to the fore (e.g. [29,30]). As a consequence of targeting visual applications, most of the reconstruction methods presented in the literature use a surface-based representation of generated objects, e.g. [31,32]. Alternatively, structural elements are represented by volumetric representations [33,34], which facilitate the extraction of model topology. Semantic enrichment of existing model geometry by indoor

Table 2
Plane fitting quality, parameters without units are dimension-less (single indoor space, c.f. Fig.3).

Plane ID	Number of points	Wall size [m^2]	Data coverage [pts/m^2]	Normalized vector v			D	Plane slope (vert.) [deg]	V error	D error	Slope error [deg]	Plane-point mean dist.[m]
				A	B	C						
1	3237	5,1	636	−0,0191	0,0069	0,9998	10,524	11,612	0,00006	0,0002	0,0003	0,0037
2	4807	5,1	944	−0,0252	0,0109	0,9996	−24,118	15,761	0,00008	0,0003	0,0003	0,0032
3	8226	9,0	910	−0,4577	0,8889	−0,0184	−10,687	91,0529	0,00003	0,0001	0,0009	0,0034
4	745	6,8	110	0,8868	0,4596	0,0491	41,834	87,1884	0,00150	0,0030	0,0772	0,0049
5	2003	9,0	222	0,4488	−0,8934	0,0201	0,8503	88,8491	0,00012	0,0003	0,0063	0,0029
6	2828	6,8	413	0,8891	0,4571	0,0232	67,937	88,6702	0,00020	0,0002	0,0066	0,0072

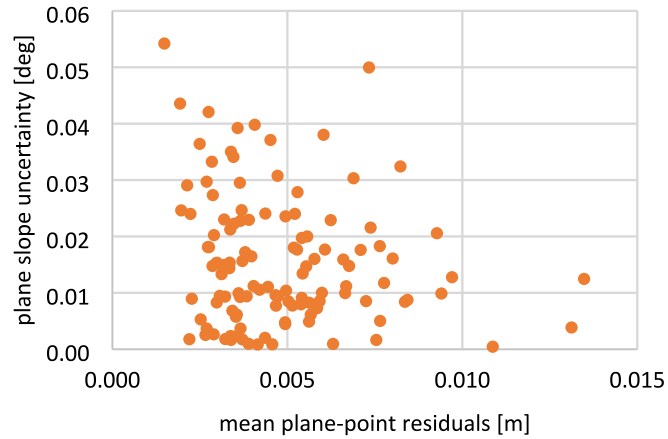

Fig. 4. Dependency between the average plane-point distance and the uncertainty of plane slope (no correlation).

objects detected from complementary data is presented in [35]. Since point clouds acquired in building interiors typically feature numerous data gaps, the increasing attention in indoor reconstruction is recently gained by 3D reconstruction based on incomplete data [36,37].

2.3. Performance evaluation

2.3.1. Assessment of 3D model accuracy

Regardless of the capabilities of data acquisition and 3D reconstruction methods, the complex indoor environment causes outliers, occlusions, data gaps, and uneven point distribution. Consequently, modeling of 3D geometry based on the collected data is a bottleneck of the whole scan-to-BIM process and contributes significantly to the accuracy of the final result [38]. The extensive research on automated solutions for the generation of as-built models is, however, not accompanied by the development of corresponding evaluation methods. Unified procedures for comparison and quality evaluation are still hardly available. Quality assessment of 3D building models is often based on visual inspection, which is subjective and time-consuming [39–41], or on the comparison with a ground truth reference [42–44]. Besides accuracy, the assessment of the geometric quality of a 3D indoor model executed by object comparison may also take into consideration two other aspects: completeness and correctness, which indicates to what extent the reconstructed elements overlay with the reference data. The accuracy aspect expresses how geometrically close the reconstructed elements are to the compared objects, assuming that the underlying geometric model is correct. Accuracy metrics used for this purpose are, however, not standardized and they are interpreted in various ways. In [8] accuracy is defined based on Euclidean distance between the 3D points sampled densely from the reference model and the closest surface in the reconstructed model. Furthermore, in [45], point-to-plane distances larger than a preset maximum value are excluded from the evaluation to avoid the influence of outliers. Besides a reference model, the reconstructed object can be also compared with the actual input data that is based on. The evaluation presented in [46] is calculated using Chamfer distance between the 3D model and the corresponding 3D points. The evaluation of the generated BIM model presented in [47] is performed by comparison with the ground truth IFC model. The experiment evaluates model recall, precision, and geometric accuracy based on the approximated distances from the point to the nearest triangulated surface on the reconstructed model.

Very often, the comparison techniques are not validated based on a common benchmark and provide a biased view of the quality of the generated model. One of the rare examples of establishing common standards and consistency in documenting existing conditions is the

Table 3
Uncertainty of 3D model vertices (single indoor space c.f. Fig. 2).

Vertrex ID	Intersection planes	Average planes' slope error [m]	Number of points	Mx [m]	My [m]	Mz [m]	error - 3d space [m]	Max displacement [m] upper 95% CI
Floor								
1	1,3,4	0,0238	12,208	0,0040	0,0021	0,0001	0,0045	0,0088
2	1,4,5	0,0302	5985	0,0036	0,0018	0,0001	0,0041	0,0090
3	1,5,6	0,0094	8068	0,0003	0,0003	0,0001	0,0005	0,0009
4	1,6,3	0,0030	14,291	0,0002	0,0001	0,0001	0,0003	0,0005
Ceiling								
5	2,3,4	0,0237	13,778	0,0005	0,0003	0,0001	0,0006	0,0011
6	2,4,5	0,0302	7555	0,0008	0,0004	0,0001	0,0009	0,0017
7	2,5,6	0,0094	9638	0,0002	0,0001	0,0001	0,0003	0,0005
8	2,6,3	0,0029	15,861	0,0002	0,0001	0,0001	0,0003	0,0006

Level of Accuracy (LOA) Specification [48] provided by the U.S. Institute of Building Documentation (USIBD). The LOA constitutes a framework for BIM-related industry to be used as guidance on the representation, modeling, and verification of as-built information. The specification refers to the acceptable tolerance range for data acquisition, as well as to the tolerance accuracy of the representation of these data in the form of a 3D model. According to the USIBD, each individual building component can be assigned to a certain accuracy level. The specification defines five geometric Levels of Accuracy in terms of Standard Deviation (Table 1), using a 95% confidence level which is

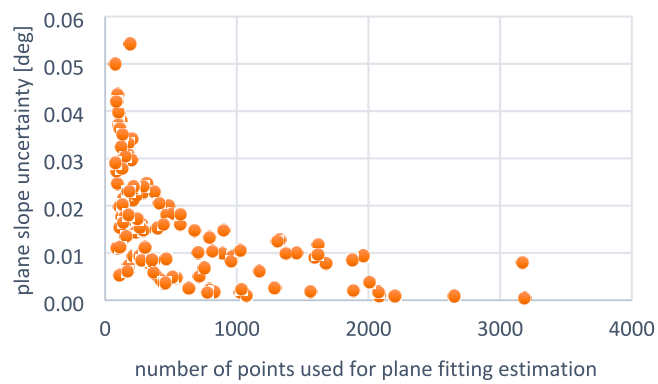

Fig. 5. Dependency between the number of points used for plane estimation and the uncertainty of plane slope.

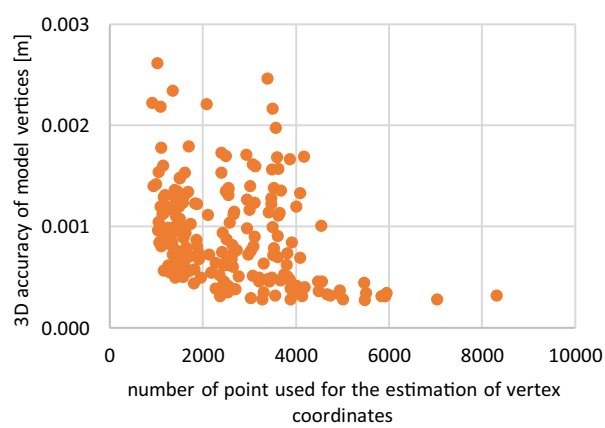

Fig. 6. Dependency between the number of points used for vertex estimation and the vertex 3D accuracy.

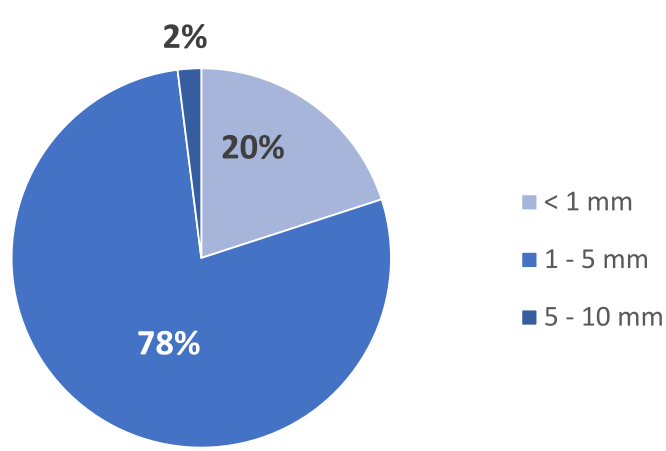

Fig. 7. Positional uncertainty of the reconstructed model vertices, specified at the 95% confidence level.

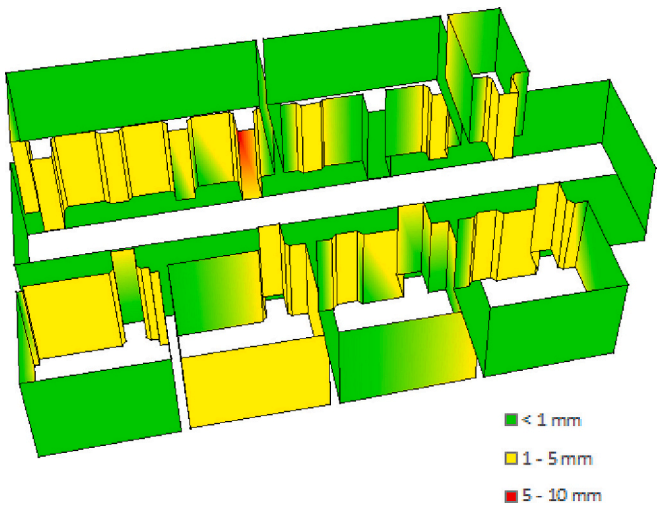

Fig. 8. Geometry uncertainty of the reconstructed indoor model, based on the uncertainty of the model vertices specified at the 95% confidence (partial transparency of the corridor is applied for better visibility).

commonly used for the estimation of positional uncertainty in surveying.

2.3.2. *Uncertainty in BIM*

A common practice in the field of 3D modeling is to represent building objects in a geometrically generalized and orthogonal fashion. Such practice works properly for as-designed BIM, which record the building design and specifications before construction begins. Since real-world conditions are hardly ever ideally orthogonal, as-built BIM regularity assumptions introduce errors. As-built models representing building after construction often differ substantially from the as-designed ones due to imprecision of the building process, reworks, unexpected site conditions, and other changes to the original design made during the construction process. As a consequence, real construction elements rarely perfectly fit the assigned parametric representation. Furthermore, in scan-to-BIM, referred as the process of creating as-built BIM from 3D point clouds, uncertainty in the digital representation is additionally increased by the inherent noise of captured 3D point clouds. Comparison-based methods for the accuracy assessment of the final modeling results assume perfect quality of the reference information. Such an assumption very often does not hold true, especially for manual reconstruction prone to subjective decisions and user experience [38]. Moreover, such methods cannot be used in a reverse way, to predict the quality of the modeling process depending on the quality of the acquired data, or even one step earlier - depending on the parameters of the acquisition sensors and geometric conditions of the captured scene. Therefore, uncertainty assessment is an important topic in the BIM-related industry, indicating the conducted process's quality. Unlike research focused on 3D reconstruction methods, this topic is, nevertheless, barely covered. Reasoning with uncertain geometry of objects is presented in [49]. Uncertain 3D building models are matched with thermal infrared images for high-quality extraction in [50]. The uncertainty of the geometry of a given BIM is a key issue in applications that requires comparison to the reference information, like for example change-detection application and monitoring of construction progress [16]. The overall accuracy of the final results is then dependent not only on the quality of the new objects being compared but – in particular – also on the reliability of the given reference model.

3. Methodology

The presented research aims at a general framework for the investigation of uncertainties in 3D building modeling. Obviously, creation of

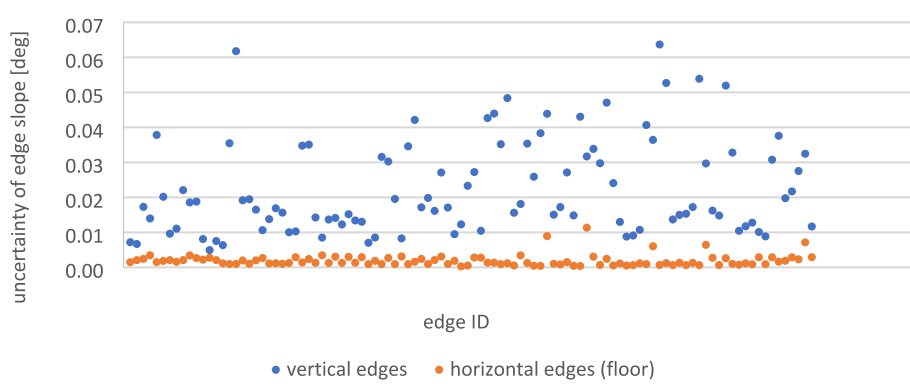

Fig. 9. Comparison of slope estimation for a different types of edges.

Table 4
Estimated inclination of 3D model edges to the horizontal plane, with corresponding uncertainties (single indoor space).

Edge type	Edge ID	Intersection planes	Number of points	Slope error [deg]	Slope - confidence intervals [deg]	
					95% CI	
vertical edges	A	3,4	8971	0,0725	86,8534	87,1377
	B	4,5	2748	0,0718	86,8083	87,0896
	C	5,6	4831	0,0065	−88,2460	−88,2207
	D	6,3	11,054	0,0051	−88,3146	−88,2944
horizontal edges - floor	E	1,3	11,463	0,0019	−0,0019	0,0057
	F	1,4	3982	0,0032	0,0338	0,0463
	G	1,5	5240	0,0019	0,0008	0,0084
	H	1,6	6065	0,0032	−0,0015	0,0110
horizontal edges - ceiling	I	2,3	13,033	0,0030	−0,0028	0,0088
	J	2,4	5552	0,0038	−0,0036	0,0112
	K	2,5	6810	0,0030	−0,0028	0,0087
	L	2,6	7635	0,0038	−0,0036	0,0111

a point cloud which is a sequence of measurement steps with individual stochastic elements (e.g. investigated in [19,21,24]), contributes to the whole chain of error propagation. As well, the generalization accuracy plays an important role in describing the quality of a reconstructed 3D model. However, in this paper we focus on a thorough analysis of geometry error propagation in the modeling process from a 3D point cloud to a 3D vectorized model. An opening step towards this goal was presented in [51]. In this preliminary investigation, we focused on the relationship between uncertainties of a subsequent step of the reconstruction process, excluding the direct influence of input data acquisition errors. In the methodological chain described below, we address this issue and propose a comprehensive error propagation model for the generation of a digital indoor scene. The presented analytical approach is also enhanced by the estimation of the probability that the reconstructed building elements fit into certain accuracy tolerance, which may also be used to verify hypotheses about its orthogonality and regularization.

Although building structural components can be abstracted in 3D space by various geometric objects, most of the presented in the literature methods for building reconstruction are plane-driven [12]. Therefore, the presented methodological pipeline (Fig. 1) starts with the investigation of error propagation in algorithms for fitting planes to the segmented 3D point cloud. Assuming that the uncertainty of the input 3D points is known (given by the manufacturer or estimated by the accuracy investigation of the acquisition sensor) we estimate the error propagation in each subsequent modeling step. All investigations consider the correlations between parameters. The estimated fitted parameters and their associating covariance matrices are used for the subsequent derivation of the building elements and their accuracies. The final result of the proposed methodological pipeline contains analytical expressions of the covariance matrices associated with the vertices and edges of the generated building model. The developed models of the building geometrical uncertainty are the basis for the following probability estimation. The error sensitivity of the implemented modeling algorithm is then thoroughly analyzed and quantified.

3.1. Estimation of error propagation

The propagation of errors in computational processes can be investigated by the methods that estimate the uncertainty of the processing results based on the input data and its associated uncertainty. This estimation can be performed stochastically or analytically. The first of the two approaches, the Monte Carlo technique, is a numerical method that works on simulated measurement data. Given some input information, a large population of corrupted data is created by repeatedly adding different random noises. This population is then processed and statistically evaluated to determine the uncertainty of the process output. Although its advantage, related to uncomplicated implementation, the Monte Carlo method provides a solution limited only to each specific problem. If the data is changed in any way, the whole simulation process has to be repeated. The second approach to uncertainty modeling is analytical and uses Taylor-series expansion of the function that describes the data processing step. To linearize inherently nonlinear problems the Taylor-series expansion is truncated to the first order. This method offers benefits of a closed-form solution, providing an analytic expression of the uncertainty of the result. Since in the presented research we aim at the establishment of a general framework for the investigation of uncertainties in 3D building modeling, the methodology presented below is focused on the analytical modeling of error propagation.

Given an explicit continuously differentiable function $f(x)$ that represents any operation on the input data and the covariance matrix Λ_x of the input data described by x, we wish to compute the covariance matrix Λ_y of the result $y = f(x)$ [53]. To achieve this, we need the Taylor series expansion of $f(x)$ around the expected value $\overline{x}$ of x:

$$f(\overline{x} + \Delta x) = f(\overline{x}) + \nabla f(\overline{x})\Delta x) + O(\|\Delta x\|)^2 \tag{3.1.1}$$

For the estimated vector y, the first order approximation to the covariance matrix Λ_y is thus given by:

$$\Lambda_y = \nabla_f \Lambda_x \nabla_f^T \tag{3.1.2}$$

where ∇f describes the Jacobian of the function, which is obtained for explicit functions $\boldsymbol{y} = f(\boldsymbol{x})$ by computing partial derivatives:

$$\nabla f = \frac{\partial f(x)}{\partial x} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \tag{3.1.3}$$

If the mathematical relation f between the input x and output y is defined as an implicit function:

$$\Phi(x, f(x)) = \Phi(x, y) = 0 \tag{3.1.4}$$

then, according to the Implicit Functions Theorem [54] the Jacobian ∇f exists if

$$det\left(\frac{\partial \Phi}{\partial y}\right) \neq 0 \tag{3.1.5}$$

and is given by:

$$\nabla_f = -\left(\frac{\partial \Phi}{\partial y}\right)^{-1} \frac{\partial \Phi}{\partial x} \tag{3.1.6}$$

3.2. Uncertainty of plane fitting

The methodological pipeline presented in this paper starts with fitting planes to the noisy 3D points. The fitting operation is used to estimate the parameters of a geometric object by the minimization of a chosen error function. The presented procedure uses a least-square method that minimalizes the sum of the squares of orthogonal distances from the points to the plane. The cost function works on the assumption that each point is subject to errors in all dimensions (instead of a single dimension only, like for example an ordinary linear regression), the whole data set can be investigated in one coordinate system, and does not require rotation or transformation to the individual coordinate system of each plane.

According to Eq. 3.1.2, the merged covariance matrix Λ_p of all points and the Jacobian ∇LLS of the linear least squares problem are needed to estimate error propagation of the fitting process. Since in terrestrial laser scanning the incidence angle affects the quality of a measured point, the uncertainty of each input point is computed individually. The matrix Λ_p is obtained by cascading the covariance matrices of all the individual points:

$$\Lambda_p = \begin{bmatrix} \Lambda_{p_1} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \Lambda_{p_m} \end{bmatrix} \tag{3.2.1}$$

A plane in 3D space is represented by the equation:

$$ax + by + cz + d = 0 \tag{3.2.2}$$

where a, b, c parameters are the components of a normal vector $\boldsymbol{n} = [a, b, c]^T$, d denotes the distance from the origin, and x, y, z describe the coordinates of a point on the plane.

A set of $N > 3$ homogenized points $\boldsymbol{p}_i = (x_i, y_i, z_i, 1)$, $i = \{1 \ldots N)$ can be arranged in a design matrix D, such that:

$$\boldsymbol{D} = \begin{bmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ \vdots & \vdots & \vdots & \vdots \\ x_N & y_N & z_N & 1 \end{bmatrix} \tag{3.2.3}$$

If the points are assumed to be coplanar, but also perturbed with noise, the following equation is satisfied:

$$D\mathrm{y} = e \tag{3.2.4}$$

where y is a vector representing the plane parameters and e is a vector of residual errors.

Considering normalized, mean-free points, the fitting process may be defined as a minimization problem of a form:

$$min_y \|D\mathbf{y}\| \text{ subject to } \|\mathbf{y}\| = 1 \tag{3.2.5}$$

The solution can be then found by solving the eigenvalue problem:

$$S\mathbf{y} = \lambda\mathbf{y} \tag{3.2.6}$$

where the scatter matrix S is defined as:

$$S = D^T D \tag{3.2.7}$$

The method delivers coefficients of the normal vector to the fitted plane. The last plane parameter, $d$is found by back substitution:

$$d = -[\bar{x}\,\bar{y}\,\bar{z}]\,[a\,b\,c]^T \tag{3.2.8}$$

Since the linear least square problem is a special case of implicit functions, eigenvalue analyses may be further used to calculate an approximation of the Jacobian of the defined minimization process. Using Eq. 3.2.6 under the assumption that the eigenvalue $\lambda_{min} \approx 0$, we implicitly define $\boldsymbol{y}$ as:

$$\Phi = S\mathbf{y} = 0 \tag{3.2.9}$$

which yields:

$$\left(\frac{\partial\,\Phi}{\partial\mathbf{y}}\right)^{-1} = S^{-1} \tag{3.2.10}$$

Considering that:

$$\frac{\partial\Phi}{\partial x} = \frac{\partial\Phi}{\partial S}\frac{\partial S}{\partial\Phi} \tag{3.2.11}$$

the Jacobian of the linear-least-squares fitting process is then given according to Eq. 3.1.6:

$$\nabla_{LLS} = -S^{-1}\frac{\partial\Phi}{\partial S}\frac{\partial \mathrm{S}}{\partial x} \tag{3.2.12}$$

In accordance with Eq. 3.1.2 and Eq. 3.2.1, the covariance matrix of the normal vector coefficients is:

$$\Lambda_n = \nabla_{LLS}\Lambda_p\nabla_{LLS}^T \tag{3.2.13}$$

The complete vector of the plane parameters is determined by the following functional relationship:

$$\mathbf{y} = f_\pi(a,b,c,\bar{x},\bar{y},\bar{z}) = [a\,b\,c - (a\bar{x} + b\bar{y} + c\bar{z})]^T \tag{3.2.14}$$

Hence, the Jacobian ∇f_π of this function yields:

$$\nabla f_\pi = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ -\bar{x} & -\bar{y} & -\bar{z} & -a & -b & -c \end{bmatrix} \tag{3.2.15}$$

Since the merged covariance matrix of the input parameters is now:

$$\Lambda_{np_m} = \begin{bmatrix} \Lambda_n & 0 \\ 0 & \Lambda_p/m \end{bmatrix} \tag{3.2.16}$$

where m is the number of input points, then the final covariance matrix of the fitted plane is computed as:

$$\Lambda_\pi = \nabla f_\pi \Lambda_{np_m} \nabla f_\pi^T \tag{3.2.17}$$

and can be directly used to extract the uncertainties of the estimated plane parameters and their correlations.

Furthermore, the vertical slope s of the extracted plane is computed based on the angle between the plane normal vector and the horizontal plane:

$$s = cos^{-1}\frac{c}{\sqrt{a^2 + b^2 + c^2}} \tag{3.2.18}$$

The related uncertainty of the plane slope is estimated taking the covariance matrix of the normal vector Λ_n and the Jacobian of the function s (Eq. 3.2.15):

$$\nabla f_s = \frac{\partial(a,b,c)}{\partial(s)} \tag{3.2.19}$$

and given by the following relation:

$$\Lambda_s = \nabla f_s \Lambda_n \nabla f_s{}^T \tag{3.2.20}$$

3.3. Positional uncertainty of model vertices

Based on the analytically derived covariance matrices of the fitted planes (Eq. 3.2.17), we investigate the error sensitivity of the reconstruction algorithm that estimates the coordinates of model vertices. Given three planes l, g, m with the covariance matrices $\Lambda_l, \Lambda_g, \Lambda_m$, the intersection point p is computed as:

$$\boldsymbol{p} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} -\left(b_l c_g d_m - b_l c_m d_g - b_g c_l d_m + b_g c_m d_l + b_m c_l d_g - b_m c_g d_l\right)/t \\ \left(a_l c_g d_m - a_l c_m d_g - a_g c_l d_m + a_g c_m d_l + a_m c_l d_g - a_m c_g d_l\right)/t \\ -\left(a_l b_g d_m - a_l b_m d_g - a_g b_l d_m + a_g b_m d_l + a_m b_l d_g - a_m b_g d_l\right)/t \end{bmatrix} \tag{3.3.1}$$

where

$$\boldsymbol{t} = a_l b_g c_m - a_l b_m c_g - a_g b_l c_m + a_g b_m c_l + a_m b_l c_g - a_m b_g c_l$$

and a, b, c, d denote the relating plane parameters of planes l, g, m, according to Eq. 3.2.2.

The covariance matrix of the data is block diagonal in the form:

$$\Lambda_{lgm} = \begin{bmatrix} \Lambda_l & 0 & 0 \\ 0 & \Lambda_g & 0 \\ 0 & 0 & \Lambda_m \end{bmatrix} \tag{3.3.2}$$

Based on Eq. 3.3.1, the Jacobian of the point coordinates function is derived by:

$$\nabla f_p = \frac{\partial\left(a_l, b_l, c_l, d_l, a_g, b_g, c_g, d_g, a_m, b_m, c_m, d_m\right)}{\partial(x,y,z)} \tag{3.3.3}$$

The X,Y,Z coordinates of the intersection point are then given by the covariance matrix described as:

$$\Lambda_p = \nabla f_p \Lambda_{lgm} \nabla f_p{}^T \tag{3.3.4}$$

3.4. Positional uncertainty of model edges

The covariance matrices of the fitted plane parameters, estimated in Section 3.2, are also used to derive a general model of error propagation for the uncertainty of the model edges. A directional vector of an intersection 3D line described by two planes l, g is computed as:

$$\boldsymbol{v} = \begin{bmatrix} v_i \\ v_j \\ v_k \end{bmatrix} = \begin{bmatrix} -b_l c_g + b_g c_l \\ a_g c_l - a_l c_g \\ a_l b_g - a_g b_l \end{bmatrix} \tag{3.4.1}$$

Based on the Jacobian of the function v:

$$\nabla f_v = \frac{\partial(a_l, b_l, c_l, a_g, b_g, c_g)}{\partial(v_i, v_j, v_k)} \tag{3.4.2}$$

and the merged covariance matrices of plane normal vectors $\Lambda_{ln}, \Lambda_{gn}$, related to a, b, c parameters, we derive the covariance matrix of the intersection line directional vector parameters by:

$$\Lambda_v = \nabla f_v \Lambda_{ln,gn} \nabla f_v{}^T \tag{3.4.3}$$

The inclination of the 3D edge of the model with respect to the horizontal plane is given by:

$$s_e = cos^{-1} \frac{v_k}{\sqrt{v_i{}^2 + v_j{}^2 + v_k{}^2}} \tag{3.4.4}$$

and its corresponding covariance matrix Λ_{s_e}, computed based on the Jacobian of s_e:

$$\nabla f_{s_e} = \frac{\partial(v_i, v_j, v_k)}{\partial(s)} \tag{3.4.5}$$

is then derived as:

$$\Lambda_s = \nabla f_{s_e} \left(\nabla f_v \Lambda_{ln,gn} \nabla f_v{}^T\right) \nabla f_{s_e}{}^T \tag{3.4.6}$$

3.5. Estimation probability

The derived covariance matrices provide us with the estimated standard deviations (SD_β) of the geometrical parameters (β), given by the corresponding square roots along the diagonal of the respective matrix. We further use this information to compute the range for the expected values of the estimated parameters. The given numbers reflect the probability of a BIM element to be contained within a certain accuracy range and enable to assign its corresponding LOA (c.f. Section 2.3). Assuming normal distribution of the errors in the fitted data, we compute confidence interval (CI_β) at the 95% confidence level which can be regarded as the standard in surveying:

$$CI_\beta = \beta \pm 1.96 * SD_\beta \tag{3.5.1}$$

4. Experiments and results

4.1. Input data

The presented methodological pipeline forms a general framework that can be used for the evaluation of the uncertainty of the plane-based 3D reconstruction process. Modeling objects can take various forms that can result from plane intersections and be described by edges and corners. The input data for the algorithm contains 3D points with estimated uncertainties, irrespective of the data collection technique and point cloud generation method. In the experiment described below, we check the suitability of the framework using exemplary data for the modeling of indoor building primary elements (walls, ceilings, floors). The test object in our study presents an indoor area covering 22 m × 15 m, located in a building of mainly educational and office use. The input data set contains coordinates of ≈ 2 million points, acquired by Leica BLK360. The scanner has a declared 3D accuracy of 6 mm at 10 m. According to the technical specification, the angular precision is equal to 40″, for both vertical and horizontal angles. The scanner has a range accuracy of 4 mm at 10 m and 7 mm at 20 m, which can be assumed to be a linear relationship throughout the entire range (0.6 m to 60 m) of the device. In the presented experiment, individual accuracy values for each 3D input point are computed based on the distance between the point and the scanner, and the related incidence angle. The developed methodological chain (c.f. Section 3) is tested on a 3D point cloud segmented into planar patches by the segmentation algorithm presented in [55]. Starting from the input set of classified wall points, the algorithm determines the best fitting planes based on the predefined criteria, using a regression method and least squares adjustment. As a result, the whole data set is grouped into segments associated with recognized 3D planes. The selection of the candidate points to the growing plane is based on the plane-point distance of 5 mm. In case of large elements like building structural parameters such thresholds enable to avoid under segmentation. Over segmentation is controlled by analyzing topological relation between neighboring segments and merging similar parallel segments. Considering that we have already roughly classified data with only single points belonging to clutters, such implementation enables to provide the segmentation results without the aforementioned errors.

The collected input data and the corresponding 3D indoor model are visualized in Fig. 2. To facilitate the presentation of the consecutive steps of the processing pipeline, additional detailed analyses are performed on a small part of exemplary data, depicted in Fig. 3. The analyses aim at the visualization of the dependency between computed numerical values of the reconstruction uncertainty and the corresponding 3D objects.

4.2. Model evaluation

4.2.1. Fitted planes

The practical application of the developed solution starts with the computation of the uncertainty of plane fitting. During the preprocessing stage, a raw point cloud is automatically segmented into planar patches. The results of this operation are depicted in Fig. 3b. The exemplary numerical statistics (related to the small model) are collected in Table 2. The numbers refer to the estimated parameters of each detected building plane (according to Eq. 3.2.2). The v error, calculated based on the parameters of the plane a,b,c, relates to the orientational part of the estimated plane, whereas the d error refers to the location part of the plane (its distance from the origin).The errors of the estimated plane parameters form the base for the subsequent derivation of the plane slopes (c.f. Eqs. 3.2.18–3.2.20).

The computed metrics also provide the number of points for each segment, the size of the corresponding wall, and the resulting point density of the wall coverage. Large differences in the number of points per plane (about 9×) are mostly related to occlusions and wall openings. Additionally, we calculate a mean absolute distance of 3D points to the estimated plane. The residuals, computed for the whole data set (Fig. 2), oscillate between 1 mm and 14 mm, with a median of 4 mm, also confirming the assumption of scanner accuracy in Section 4.1. The values are compared against the uncertainties of plane slope estimation in Fig. 4.

Although mean point-plane residuals are often applied in 3D modeling field as an indicator for plane-fitting quality, in fact, there is no direct correlation between these two indicators. The same lack of correlation is confirmed in the visual analysis of the small data model (Fig. 2). For example, the point segment assigned with plane *6* reveals large point-plane residuals, while its uncertainty of plane fitting, remains relatively small. It is also visible that the plane fitting process achieved the highest uncertainty for the segment *4*, which is the wall with the worst data coverage. The same dependencies are observed in the large model. The inverse dependency between the number of wall points and fitting uncertainty is illustrated in Fig. 4. Besides data coverage, also the overall shape of the plane affects the fitting quality, leading to higher uncertainty related to the planes with large height-to-length ratio. The accuracy of the estimated plane slope varies from 0.0004 deg. up to 0.0542 deg. and is inversely correlated with the number of the corresponding 3D points used for the estimation.

4.2.2. Model vertices

In the following step of 3D model evaluation, we investigate the sensitivity of the reconstructed model vertices to the plane fitting uncertainties. The exemplary values of the small model referring to the 3D vertex accuracy (maximum point displacements in 3D space), together with the information related to the corresponding intersection planes are presented in Table 3. We can observe that point errors related to z coordinate are nearly constant for all the points, while the planar uncertainty varies. The direct impact of the number of points used for the estimation of a 3D vertex on its uncertainty (presented in Fig. 6) is not observed anymore, contrary to the previous step of plane fitting (c.f. Fig. 5).

The largest 3D errors are observed for the bottom vertices created by the intersection with *plane 4,* which has the highest fitting uncertainty and no points in the lower segment part. Although plane fitting errors propagate into the errors of the 3D vertices, the final uncertainty of the result is also affected by the geometrical configuration of the collected data. The intersection of planes with large average slope errors does not directly lead to large uncertainty of the resulting vertex. In the visualization of the results, we can notice a clear dependency between vertex neighborhood visibility and vertex uncertainty. The average 3D positional error of the whole data set, based on 208 intersection points, is equal to 1.0 mm, with values ranging from 0.3 mm to 2.6 mm.

Using the standard error analytically derived from the covariance matrix of the intersection point coordinates, we can determine confidence regions for the estimation of positional uncertainty (Section 3.5). For the 3D indoor model reconstructed in our experiment (presented in Fig. 2), the 3D positional uncertainty of the vertex coordinates determined at the 95% confidence level falls within the range of 0.5–5.1 mm, with the average value of 2.0 mm. For almost all of the vertices (98%, c.f. Fig. 7) the estimated values are smaller than 5 mm. In terms of USIBD specification (Table 1), the reconstructed building elements of the whole model can be therefore assigned to the accuracy levels from LOA30 up to LOA50. The geometry uncertainty of the reconstructed 3D model is visualized in Fig. 8.

4.2.3. Model edges

In the last numerical experiment, we assess how the plane fitting errors affect the positional uncertainty of model edges generated by the intersection of two planes. Based on the estimated parameters of 312 edges – 3D line directional vectors and the associated covariance matrices – we derived the slope of the model edge, computed as an inclination to the horizontal plane, as well as its corresponding uncertainty (exemplary values for the small data set are collected in Table 4). The largest uncertainties are observed for the edge A and edge B, which are generated by the intersection with *plane 4*, with the worst fitting quality. The impact of input point geometry on the uncertainty of the reconstructed edges is observed again, but its effect is lower than in the case of vertex estimation. This is, for example, visible for the edge H, which is nearly completely occluded however its quality measures are comparable to the results computed for the very well captured ceiling edges. The mean slope error of all the model edges is equal to 0.0094°, however, the results significantly differ between vertical edges (average errors 0.0225°) and horizontal ones (0.0028°). The results are compared and depicted in Fig. 9. In almost all the cases, the orthogonality of a building model, commonly achieved in the building reconstruction by the forced model regularization, is not indicated by the computed values of edge slopes and their uncertainties. Because of a common practice in the 3D modeling field to enforce the orthogonal representation of the generated models, we investigate the probability that the reconstructed edges indeed fulfill the orthogonality constraints. Level of confidence measures are computed for the entire dataset using an analytically determined covariance matrix of the edge slope and estimating upper and lower confidence limits. For the 3D indoor model reconstructed in our experiment, the results enable to interfere with 95% probability that only 2% of the reconstructed edges could be indeed considered orthogonal.

5. Conclusion

The building modeling field lacks research investigating the impact of the reconstruction procedure on the accuracy of generated BIM, concerning given uncertainties of the data. This study fills this gap by establishing a general framework for the investigation of uncertainties in 3D reconstruction. The paper gives an insight into error propagation analysis for the estimation of building model geometric parameters, based on the Taylor series approximation. The novelty of the presented approach results from the consideration of statistical uncertainties in the modeling process. This allows for internal quality analyses of the vectorized models, without a need for reference data. Furthermore, we show how to apply the computed uncertainties to infer a final accuracy assessment based on the Level of Accuracy specification for BIM. The probability of the reconstructed building elements to belong to certain tolerance thresholds is quantified, which enables to assign the elements with the related LOA.

The presented workflow is suited for BIM objects wherein shapes can be determined by planes and their intersections. It may be extended to object representations including non-planar geometric primitives such as cylinders, hemispheres or also NURBS. Each consecutive step of 3D reconstruction is related to the derived error propagation model. Taking as an input segmented 3D points, we analytically estimate the covariance matrix of the fitted plane parameters and investigate the uncertainties associated with the reconstructed model elements computed based on them. The influence of input errors on the uncertainty of the reconstruction results is thoroughly studied and quantified. We also investigate the impact of data density and spatial configuration on the accuracy of the reconstructed elements. The primary aim of the research is to develop a general model for the evaluation of the uncertainty of the 3D reconstruction process, regardless of the modeling object type. For the demonstration of a practical solution, the developed procedure was applied to an exemplary indoor 3D point cloud. For the virtual model automatically generated from the input data, the analyses reveal the average positional error of the 3D vertex equals 1.0 mm. Expected positional uncertainties of the model vertices determined at the 95% confidence level range from 0.5 to 5.1 mm. In terms of LOA specification, 78% of the reconstructed vertices can be assigned to LOA40. The mean error of the inclination of the model 3D edges to the horizontal plane is equal 0.0094°. For 98% of the edges, slope values of 0° or 90° commonly used for building model regularization, are out of the confidence levels computed at the 95% probability level. Assuming the reliability of the scanner self-leveling, such results would question the appropriateness of the commonly used orthogonality constraints if real as-build conditions are the goal of the modeling process.

The underlying methodology so far neglects unflatness of a surface assuming that the planes being reconstructed are exactly planar. In future work, this hypothesis will be verified using for instance residuals from the input data. The work also forms a basis to test the hypothesis on whether it is justified to impose regularity rules on the reconstructed elements of the model. Furthermore, follow-up work will take under consideration systematic (and thus correlated) errors related to the calibration of the terrestrial scanner and investigate alternative sources for 3D point cloud acquisition (e.g. personal laser scanners, or smartphones equipped with lidar sensors). In the presented practical part of the experiment, we demonstrate the suitability of the developed framework for the modeling of indoor building primary elements. We plan to enhance the scope of the investigation using BIM developed for a

very precise representation of building features with high detailing requirements.

Declaration of Competing Interest

Malgorzata Jarzabek-Rychard reports financial support was provided by Polish National Agency for Academic Exchange.

Data availability

Data will be made available on request.

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